

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 6

Total Marks:30

Annexure A

Scenario-Based

Code of Conduct:

*I (K. Deepak / 192320078) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

1. Use Case Diagram Scenario – Smart Pharmacy Management System

System Analysis

The Smart Pharmacy Management System allows customers to search for medicines, upload prescriptions, place orders, and make online payments.

Pharmacists verify prescriptions and dispense medicines, while administrators manage inventory and user accounts.

Actors

- Customer
- Pharmacist
- Administrator
- Payment Gateway (External Actor)

Use Cases

Customer

- Register/Login
- Search Medicines
- Upload Prescription
- Place Order
- Make Payment
- Track Order

Pharmacist

- Verify Prescription
- Dispense Medicines
- Update Order Status

Administrator

- Manage Medicine Inventory
- Manage User Accounts
- Generate Reports

Payment Gateway

- Process Payment
- Verify Payment

Relationships

<<include>>

- Place Order includes Search Medicines.
- Place Order includes Make Payment.

- Make Payment includes Process Payment.

<<extend>>

- Upload Prescription extends Place Order (only for prescription medicines).
- Verify Prescription extends Upload Prescription.

Generalization

- Customer can be specialized as:
 - Regular Customer
 - Premium Customer

2. Class Diagram Scenario – Smart Hotel Management System

Classes

Guest

Attributes

- guestID
- name
- phone
- email

Methods

- bookRoom()
- cancelReservation()

Room

Attributes

- roomNo
- roomType
- price
- status

Methods

- checkAvailability()
- updateStatus()

Reservation

Attributes

- reservationID
- checkInDate
- checkOutDate
- reservationStatus

Methods

- createReservation()
- cancelReservation()

Payment

Attributes

- paymentID
- amount
- paymentDate
- paymentStatus

Methods

- processPayment()
- generateReceipt()

Receptionist

Attributes

- employeeID
- name
- shift

Methods

- assignRoom()
- checkInGuest()

Relationships

- Guest associates with Reservation (1 Guest can have many Reservations).
- Reservation associates with Room (one Reservation is for one Room).
- Reservation composes Payment (a reservation has one payment).
- Receptionist associates with Reservation (manages reservations).
- Hotel aggregates Rooms and Receptionists (if Hotel class is included).
- Guest may inherit from a common superclass Person.

Multiplicity

- Guest (1) → Reservation (0..*)
- Reservation (1) → Room (1)
- Reservation (1) → Payment (1)
- Receptionist (1) → Reservation (0..*)
-

3. Sequence Diagram Scenario – Online Banking System

Participating Objects

- Customer
- Authentication Service
- Banking Server
- Database
- OTP Service
- Notification Service

Sequence of Operations

1. Customer enters login credentials.
2. Authentication Service validates credentials with the Database.
3. Database returns authentication result.
4. Customer selects Fund Transfer.
5. Banking Server receives transfer request.
6. Banking Server checks account balance from the Database.
7. Database returns balance details.
8. Banking Server requests OTP from the OTP Service.
9. OTP Service sends OTP to the customer.
10. Customer enters the OTP.
11. Banking Server verifies the OTP.
12. Banking Server updates the transaction in the Database.
13. Database confirms successful transaction.
14. Banking Server sends confirmation to the Notification Service.
15. Notification Service sends SMS/Email confirmation to the customer.

4. Activity Diagram Scenario – Passport Application System

Workflow

1. Start.
2. Submit passport application.
3. Upload supporting documents.
4. Fork into two parallel activities:
 - Identity verification.
 - Document verification.
5. Join after both verifications are completed.
6. Decision Node:
 - If verification is approved, proceed to fee payment.
 - If verification is rejected, notify the applicant and end the process.

7. Applicant pays processing fee.
8. Passport is processed and printed.
9. Passport is dispatched.
10. Applicant receives the passport.
11. End.

Decision Points

- Verification Approved?
- Payment Successful?

Parallel Activities

- Identity verification.
- Document verification.

5. State Diagram Scenario – Smart Elevator Control System

States

- Initial State
- Idle
- Door Open
- Door Closed
- Moving Up
- Moving Down
- Emergency Stop
- Maintenance Mode
- Final State

Transitions

- System starts in the Idle state.
- Elevator call opens the door (Door Open).
- Door closes (Door Closed) after passengers enter or exit.
- If the destination is above, transition to Moving Up.
- If the destination is below, transition to Moving Down.
- On reaching the requested floor, transition to Door Open.
- After passengers leave, transition to Door Closed, then Idle.
- From any state, pressing the emergency button transitions to Emergency Stop.
- After the emergency is cleared, return to Idle.
- A maintenance request from any state transitions to Maintenance Mode.
- After maintenance is completed, return to Idle.

Triggering Events

- Floor request
- Door open/close command
- Arrival at destination
- Emergency button pressed
- Emergency cleared
- Maintenance request
- Maintenance completed

6. Deployment Diagram Scenario – Smart Home Automation System

Hardware Nodes

- Smart Sensors
- IoT Controller
- Home Gateway
- Cloud Server
- Database Server
- Mobile Device

Software Artifacts

- Sensor Firmware
- IoT Controller Software
- Home Automation Application
- Cloud Management Software
- Database
- Mobile Application

Communication Links

- Smart Sensors communicate with the IoT Controller using ZigBee, Wi-Fi, or Bluetooth.
- IoT Controller sends data to the Home Gateway.
- Home Gateway communicates with the Cloud Server through the Internet.
- Cloud Server stores and retrieves data from the Database Server.
- Mobile Application connects to the Cloud Server to monitor and control home devices remotely.
- Cloud Server sends notifications and control commands back to the Home Gateway and IoT devices.

Deployment Summary

The Smart Home Automation System consists of smart sensors connected to IoT controllers within

the home. The controllers communicate through a home gateway to cloud servers, where data is processed and stored in a database. Users interact with the system through a mobile application, enabling remote monitoring and control of lighting, security systems, and household appliances.

RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand